

TFT-GCN: A Time-Frequency Based Model for Time Series Anomaly Detection

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Abstract—Time series anomaly detection is essential for maintaining the stability and operational efficiency of industrial systems. However, most existing approaches emphasize temporal dependency modeling while paying limited attention to spectral characteristics embedded in the data. This limitation prevents models from effectively capturing both localized patterns and global trends, thereby reducing the accuracy of anomaly detection. To overcome these challenges, we introduce TFT-GCN, a novel anomaly detection model that integrates temporal and spectral analysis modules to simultaneously extract local and global information. Specifically, in the frequency-domain branch, we adapt the attention mechanism to extract critical information about frequency combinations, enabling the efficient extraction of global trends. In the time-domain branch, we introduce a cross-variable graph convolutional network, which not only models both homogeneous and heterogeneous interactions between series but also increases resilience to noise. Additionally, we incorporate a multi-scale attention mechanism that efficiently captures local dependencies across varying time scales to prevent the model from focusing too much on features at a single time scale. By combining the complementary strengths of temporal and spectral analyses, TFT-GCN offers a powerful framework for identifying anomalies. Experimental results on seven benchmark datasets verify that TFT-GCN surpasses most existing methods, highlighting its superior performance and robustness in time series anomaly detection.

Index Terms—Time series anomaly detection, time domain, frequency domain.

I. INTRODUCTION

MANY large-scale industrial systems rely on the continuous generation and processing of multivariate time series data to ensure efficient operation. These data are essential for monitoring system health and optimizing performance. However, anomalies inevitably occur in these systems, often resulting

in equipment failures, production bottlenecks, process deviations, or safety hazards. The detection of industrial anomalies is complicated by several factors, including high-dimensional sensor observations, intricate correlations among variables, and continuously evolving industrial dynamics. Consequently, considerable attention has been devoted to industrial time series anomaly detection to enhance operational efficiency and prevent costly system failures [1], [2].

However, in most real-world datasets, data points predominantly fall within normal ranges, with anomalies being rare occurrences. In addition, manual labeling of anomalies takes a lot of time and effort. Normal patterns may change dynamically over time, and the definition of anomalies often relies on specific contexts, which makes it even more difficult to define anomalies accurately [3]. Under such conditions, methods depending on labeled samples are often difficult to apply effectively because annotated anomaly data are extremely limited. As a result, increasing attention has been directed toward unsupervised anomaly detection approaches. The core of such methods lies in learning normal data patterns and identifying anomalies by means of reconstruction error or prediction bias, for example. Recent studies have shown that unsupervised approaches can achieve competitive performance across diverse application scenarios [4], [5], [6].

However, there are still challenges in detecting anomalies within time series. Firstly, time series data exhibit global and local dependencies, which are intertwined, complicating the modeling of time series features. Representative Transformer architectures, including Autoformer [7] and FEDformer [8], effectively capture long-range temporal dependencies but mainly operate in the time domain. Consequently, their robustness to high-frequency noise remains limited, and global spectral information cannot be sufficiently characterized. Models like PatchTST [9] and TimesNet [10] focus more on local fragment-level or two-dimensional representations, which effectively capture patterns within cycles. Although these methods demonstrate strong capabilities in modeling temporal characteristics across multiple scales, they still remain vulnerable to noise interference.

In contrast, frequency-domain analysis provides a complementary perspective for capturing global periodic structures and spectral regularities. Representing time-series signals in the frequency domain enables the extraction of long-range dependencies and periodic structures that are not easily captured through temporal observations alone [11]. In general,

Received 30 November 2025; revised 31 May 2026; accepted 13 June 2026. Date of publication 29 June 2026; date of current version 8 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62441237, Grant U25A20435, Grant 62372194, Grant 62272189, and Grant 62272357 and in part by Wuhan City Joint Innovation Laboratory for Next-Generation Wireless Communication Industry Featuring Satellite-Terrestrial Integration under Grant 4050902040448. Recommended for acceptance by J. Tang. (*Corresponding author: Bolong Zheng.*)

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Our code is available at <https://github.com/xxxdongjiu/TFT-GCN>.

Digital Object Identifier 10.1109/TKDE.2026.3707947

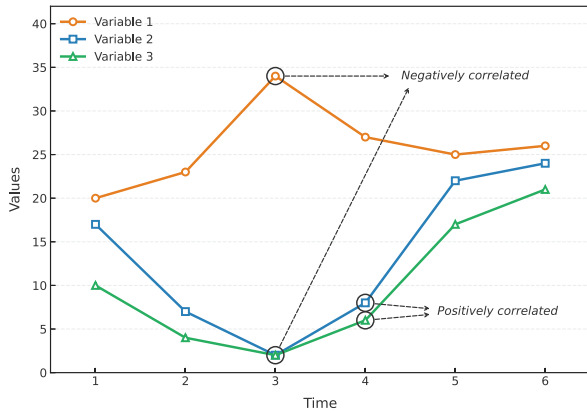


Fig. 1. Homogeneous and heterogeneous relationships between variables.

low-frequency components reflect the underlying periodic behavior of the system, while high-frequency components are more related to noise or irregular fluctuations. Therefore, frequency-domain analysis can suppress noise interference and emphasize dominant periodic patterns, making it particularly suitable for seasonal or highly periodic time series. More importantly, frequency-domain modeling is naturally aligned with anomaly characteristics. Under normal conditions, industrial systems usually exhibit relatively stable periodic rhythms, leading to concentrated spectral energy distributions. When anomalies occur, these regular patterns are disrupted, often causing spectral energy shifts or abnormal frequency responses. As a result, frequency-domain analysis can effectively identify structural deviations and periodic disruptions that may be less distinguishable in the time domain. However, it is less sensitive to non-periodic variations. Therefore, combining temporal dependencies with spectral patterns helps construct richer feature representations for time series anomaly detection.

Finally, in multivariate settings, inter-variable interactions further complicate the extraction of normal patterns from the data. As shown in Fig. 1, the interactions among variables may contain both consistent correlations and heterogeneous dependency patterns. The Anomaly Transformer [12] introduces association discrepancy as a criterion for anomaly detection, yet it still lacks the ability to robustly capture heterogeneous inter-variable correlations. Although FuSAGNet [13] employs GNNs to capture relationships between variables, it adopts a unified modeling approach that does not account for the homogeneous and heterogeneous relationships among variables. In fact, these two types of relationships contribute to the invariant associations within the time series. Learning these invariant associations, which capture both homogeneous and heterogeneous relationships, not only enables effective modeling of the internal relationships but also enhances robustness against noise [14]. Based on the above analysis, there are three challenges that need to be addressed: (1) how to effectively integrate local and global dependencies; (2) how to extract and leverage informative frequency-domain representations; and (3) how to learn invariant associations between series.

To tackle these challenges, we propose TFT-GCN, incorporating a temporal branch and a frequency branch to capture both

fine-grained fluctuations and broader trends within time series data. In the temporal branch, we utilize Cross-Variable Graph Convolutional Network (GCN) to learn invariant relationships, effectively capturing both homogeneous and heterogeneous interactions across variables. In addition, to model short-term correlations under multiple temporal granularities, we introduce a multi-scale attention module. Specifically, this module performs spectral analysis on the input data through Fast Fourier Transform (FFT). After extracting primary periodic scales, the system employs multi-head attention to discern and integrate local features at various time scales. In the frequency branch, we prioritize the extraction of global patterns from frequency combinations. Analyzing time series from the frequency domain perspective involves transforming data from the real number field to the complex number field, making the proper handling of complex numbers a crucial aspect. Approaches that treat complex numbers as real numbers by separating their real and imaginary parts often struggle to capture the intricate interactions within complex data, potentially leading to biased analysis results. Therefore, we adapt the attention mechanism to process frequency combinations directly from the complex domain. Firstly, shifting the time data to the spectral domain enables the system to identify and extract relevant frequency-based characteristics. Then, the attention mechanism operates within the complex domain to analyze spectral features and capture global patterns. To capture dual-domain features more efficiently, we integrate reconstruction errors from temporal and spectral representations into a unified loss function during training. This training strategy effectively guides the model to learn essential features from both domains, which improves the model's capability to characterize complicated temporal dynamics. Overall, TFT-GCN fully utilizes the complementary strengths of analyzing temporal patterns and spectral features, enabling it to identify local and global dependencies within series and model dynamic correlations between series, thereby improving the performance and robustness in detecting anomalies. The main innovations of our work are as follows:

- To better model short-term variations across multiple time horizons, the temporal branch employs a multi-scale attention architecture. In this way, informative temporal structures at multiple scales can be selectively emphasized.
- To exploit informative frequency-domain representations, a frequency-domain attention module is developed to highlight important frequency interactions and dominant spectral patterns. This significantly strengthens the model's capacity for spectral feature extraction and global contextual understanding.
- To learn invariant cross-variable associations, we incorporate a cross-variable Graph Convolutional Network (GCN) into the time-domain pathway. By modeling inter-series dependencies as a learned graph structure, the model captures high-level relational semantics, improving robustness against noise.
- Evaluation results on seven public benchmark datasets indicate that the proposed method consistently outperforms existing approaches in anomaly detection accuracy.

The subsequent sections are arranged as follows. Section II introduces relevant studies of anomaly detection. Then, Section III introduces the proposed TFT-GCN method, detailing its architecture and network design. Section IV provides an empirical evaluation of TFT-GCN on multivariate time series datasets and compares its performance with leading methods. Additional analyses are further performed to investigate the effectiveness of different model components. The paper concludes with Section V, summarizing the outcomes and insights.

II. RELATED WORK

A. Modeling of Intra-Series Correlation

The internal correlation of time series consists of local and global dependencies. RAMEL [15] proposed using a set of decoders with varying decoding lengths to capture multi-resolution temporal information, addressing the inherent multi-periodic characteristics in time series data. OmniAnomaly [16] efficiently models the typical behavior of multivariate time series by combining gated recurrent units (GRUs) with a random-variable concatenation technique. P. Soda et al. [17] proposed a novel symbolization method that simultaneously considers inter-variable dependencies and temporal variations, converting multivariate sequences into symbolic forms that preserve richer structural information while reducing representation redundancy, thereby enhancing the performance of the model. The Transformer architecture has become widely adopted for detecting anomalies owing to its proficiency in modeling long-range temporal relationships. Among these methods, Anomaly Transformer [12] introduces association discrepancy as a criterion for anomaly detection. D3R [18] dynamically separates time series into stable and trending components using a hybrid data-time attention mechanism, mitigating the limitations of local sliding windows. Similarly, Koopa [19] leverages Koopman operator theory to decouple time-invariant and time-varying components within a series, enabling robust modeling of non-stationary dynamics with improved efficiency. Although these methods capture features of time series, they primarily model the data from a time-domain perspective, making them susceptible to noise interference and challenging in terms of integrating local and global information efficiently. In contrast, while TimesNet [10] extracts primary periodic features using the Fast Fourier Transform, its model architecture is rooted in pure computer vision, which may not be fully optimal for time series data.

B. Modeling of Inter-Series Correlation

Modeling inter-series relationships is essential for effective anomaly detection. GDN [20] leverages its modeling capabilities to capture complex relationships between sensors, emphasizing inter-series dependencies and effectively addressing irregularities in complex temporal data. As awareness of these relationships has grown, numerous studies have increasingly employed various methods to capture and model them. For instance, FuSAGNet [13] captures complex interdependencies in high-dimensional sensor data by jointly optimizing sparse autoencoder (SAE) reconstruction and GNN-based prediction,

while DVGCRN [21] combines a variational graph convolutional recurrent network (VGCRN) with a probabilistic generative network guided by deep embeddings (DEPN) to effectively characterize coupled spatial-temporal correlations embedded in multivariate sequences. McSE [22], a novel predictability metric, integrates multivariate constraint relationships to significantly enhance the upper limit of predictability for multivariate time series forecasting. Additionally, iTransformer [23] innovatively reconfigures the traditional architecture by employing self-attention mechanisms to model inter-variable relationships and utilizing feed-forward layers to extract temporal patterns within individual variables, significantly enhancing the representation of time series data. FourierGNN [24] constructs a hyper-variable graph and applies a Fourier graph operator to efficiently capture cross-variable dependencies in the spectral domain, achieving superior accuracy and efficiency. REPST [25] introduces a spatio-temporal reprogramming framework that decouples temporal and spatial signals in the frequency domain to align time series with pre-trained language models, demonstrating strong adaptability under data scarcity. Similarly, DSTF [26] separates diffusion and intrinsic traffic signals, with D2STGNN further leveraging dynamic graph learning to capture spatio-temporal correlations more effectively. Furthermore, ModernTCN [27] applies the concept of decoupling to perform independent convolution operations along temporal, feature, and variable axes. However, the relationships between variables are typically dynamic and prone to noise interference. Building on previous work [14], we strengthen the model's resilience to noise by learning invariant relationships that capture both homogeneous and heterogeneous interactions across variables.

III. METHODS

This paper defines the input as $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\} \in \mathbb{R}^{T \times D}$, with T denoting the sequence length and D representing the feature dimension. Each time point carries a ground truth label y , where $(y = 0)$ signifies normal and $(y = 1)$ denotes anomalous. Our task is to generate anomaly labels for all time points, denoted as $\hat{\mathbf{y}} = \{\hat{y}_1, \dots, \hat{y}_T\}$, where $\hat{y}_t \in \{0, 1\}$, with the objective of matching the ground truth labels as closely as possible.

A. Overall Architecture

As previously discussed, our study aims to enhance anomaly detection methods by strengthening the extraction of local and global patterns. To this end, we propose TFT-GCN, which exploits the complementary advantages of temporal and spectral analyses. Fig. 2 illustrates the TFT-GCN framework, comprising a temporal branch and a frequency branch. The temporal branch integrates a cross-variable graph convolutional network (GCN) to model interdependencies among series with a multi-scale attention mechanism for extracting local features across different scales. Conversely, the frequency-domain branch utilizes a complex-valued spectral attention mechanism to extract global information from frequency components. Sections III-B and III-C detail the temporal and frequency branches, respectively, while Algorithm 1 summarizes the core steps of the approach.

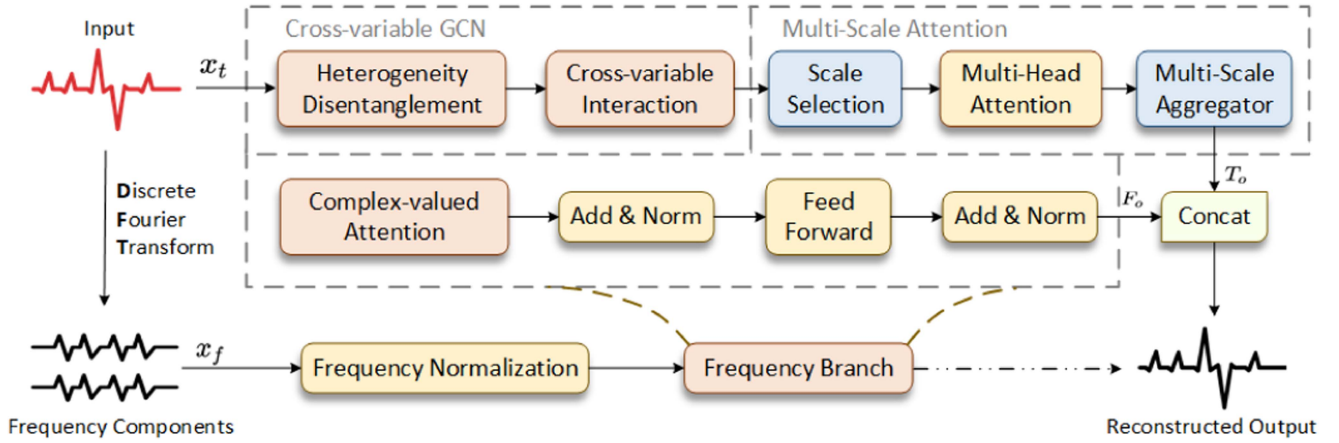


Fig. 2. Overview of the proposed TFT-GCN framework for time series anomaly detection, comprising three main components: the cross-variable GCN, which learns invariant relationships that capture both homogeneous and heterogeneous interactions across variables; the multi-scale attention mechanism, which captures local features at different temporal scales; and the complex-valued spectrum attention mechanism, which analyzes frequency components to learn spectral features and global patterns of the series.

B. Temporal Branch

1) *Cross-Variable GCN*: In multivariate time series, the relationships among variables can provide critical insights for anomaly detection. To capture these inter-variable relationships, we utilize the widely used Graph Convolutional Network (GCN). Since GCN operates on a graph structure, we first perform graph structure learning. Specifically, the graph construction regards each variable as a node, where edge connections reflect inter-series relationships. We define two learnable vectors $vec_1 \in \mathbb{R}^{D \times h}$ and $vec_2 \in \mathbb{R}^{D \times h}$ to initialize the adjacency matrix according to the following formula:

$$\mathbf{E} = \text{SoftMax} \left(\text{ReLU} \left(vec_1 (vec_2)^T \right) \right) \quad (1)$$

where the $\text{ReLU}()$ is the activation function and it is used to regularize the adjacency matrix, ensuring that the weights are non-negative. Subsequently, $\text{SoftMax}()$ is applied to normalize the node weights, guaranteeing that the total weight assigned to connected nodes sums to one.

Variable relationships in multivariate time series naturally encompass both homogeneous and heterogeneous dynamics. Traditional graph convolutional networks (GCNs) typically model these relationships uniformly, which can lead to redundant variable interactions and high susceptibility to background noise. To address these issues, we utilize a Cross-Variable GCN designed to explicitly decouple inter-variable correlations into two categories. The fundamental rationale behind this design is rooted in the underlying physical dynamics of complex systems. Specifically, for each node, the nodes with the n highest correlation weights are regarded as having homogeneous relationships with it. Physically, this high alignment indicates that these variables are co-driven by shared underlying mechanisms or exhibit synchronous seasonal patterns. Aggregating features from these Top- n neighbors provides mutually consistent representations, thereby facilitating reliable characterization of normal system dynamics. Conversely, the nodes with the n lowest correlation weights are considered to have heterogeneous relationships. Rather than merely representing “independent”

variables, extreme dissimilarity in dynamic systems typically signifies complementary, competitive, or out-of-phase regulatory mechanisms (e.g., heating load versus ambient temperature). By isolating these Bottom- n nodes, the model explicitly captures these “adversarial” or negative correlations, which act as strict operational constraints defining the system’s normal manifold. Consequently, when anomalies disrupt these intricate structural dependencies, the model fails to reconstruct the input accurately, yielding a prominent reconstruction error. Furthermore, by discarding the intermediate weights—which generally represent coincidental or spurious statistical overlaps—this bipartite Top/Bottom- n routing naturally acts as a topological filter, effectively decoupling structural noise and ensuring that only the most physically meaningful interactions are preserved for representation learning. For a node v_i , its two sets of neighbor nodes are denoted as:

$$\mathcal{N}_{neg}(v_i) = \text{Bottom-}N_{neg}(\mathbf{E}(v_i)) \quad (2)$$

$$\mathcal{N}_{pos}(v_i) = \text{Top-}N_{pos}(\mathbf{E}(v_i)) \quad (3)$$

The weights for these two types of relationships are computed as follows:

$$\mathbf{E}[i, j] = \begin{cases} -\frac{\frac{1}{\mathbf{E}[i, j]}}{\sum_{v_n \in \mathcal{N}_{neg}(v_i)} \frac{1}{\mathbf{E}[i, n]}}, & v_j \in \mathcal{N}_{neg}(v_i), \\ \frac{\mathbf{E}[i, j]}{\sum_{v_n \in \mathcal{N}_{pos}(v_i)} \mathbf{E}[i, n]}, & v_j \in \mathcal{N}_{pos}(v_i), \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The mathematical operations in this formulation are directly tailored to reflect their respective physical significance. For homogeneous edges, a standard proportional normalization is applied, as higher initial correlation scores naturally translate to stronger positive reinforcement. For heterogeneous edges, the operations are highly non-trivial. Since these nodes are selected precisely because their initial correlation scores ($\mathbf{E}$) are extremely small (approaching zero), applying the reciprocal ($1/\mathbf{E}$) mathematically inverts the scale, thereby assigning the largest absolute weight magnitudes to the most severely dissimilar variables. Furthermore, the negative sign explicitly encodes

these features as oppositional signals, maintaining the system's complementary constraints during aggregation. Finally, forcing intermediate edges to 0 filters out nodes with weak relationships to the target node, specifically those that lack strong homogeneous or heterogeneous associations.

Subsequently, we aggregate information using the updated weight matrix, as described by the following formula:

$$\mathbf{X}_g^{(l)} = \sigma \left(\mathbf{E} \mathbf{X}_g^{(l-1)} \mathbf{W}^{(l)} \right) \quad (5)$$

Here, $\mathbf{X}_g^{(l)} \in \mathbb{R}^{T \times D}$ corresponds to the feature representation at layer l , where $\mathbf{X}_g^{(0)}$ is the regularized form of $\mathbf{X}$, denoted as $\hat{\mathbf{X}} \in \mathbb{R}^{T \times D}$, which is obtained through the RevIN process. $\mathbf{W}^{(l)}$ refers to the trainable matrix for layer l .

2) *Multi-Scale Attention*: The temporal branch focuses on extracting local patterns from time series data. However, due to significant variations in characteristics across different scales, not all scales contribute equally. To effectively extract the most critical local features, we introduce a multi-scale attention mechanism.

To begin, the tensor capturing the relationships among series is transformed from D dimensions into a d_{model} -dimensional space. This transformation, represented as $\mathbf{X}_g \rightarrow \mathbf{X}_{\text{emb}} \in \mathbb{R}^{T \times d_{\text{model}}}$ is achieved using the following formula:

$$\mathbf{X}_{\text{emb}} = \text{Conv1D}(\mathbf{X}_g) + \text{PE} \quad (6)$$

where $\text{Conv1D}()$ represents a one-dimensional convolutional operation. It is designed to map D -dimensional data to d_{model} dimensions. Additionally, $\text{PE} \in \mathbb{R}^{T \times d_{\text{model}}}$ refers to positional encoding.

The multi-scale attention module operates residually. At layer l , the input $\mathbf{X}^{(l-1)} \in \mathbb{R}^{T \times d_{\text{model}}}$ undergoes processing as formally represented by the following equation:

$$\mathbf{X}^{(l)} = \text{Multi-ScaleBlock} \left(\mathbf{X}^{(l-1)} \right) + \mathbf{X}^{(l-1)} \quad (7)$$

where $\text{Multi-ScaleBlock}()$ defines the core function of the multi-scale attention. Specifically, $\mathbf{X}^{(0)} = \mathbf{X}_{\text{emb}}$. The time branch concludes with a linear projection that transforms the multi-scale attention module's output, $\mathbf{X}^{(L)} \in \mathbb{R}^{T \times d_{\text{model}}}$, into $\mathbf{X}_T \in \mathbb{R}^{T \times d_{\text{model}}}$. This process is formally expressed as:

$$\mathbf{X}_T = \mathbf{X}^{(L)} \mathbf{W} + \mathbf{b} \quad (8)$$

where $\mathbf{W} \in \mathbb{R}^{d_{\text{model}} \times D}$ is a trainable weight matrix, and $\mathbf{b}$ represents the bias term. The resulting $\mathbf{X}_T$ serves as the output from the time branch.

In the l -th layer, the multi-scale attention employs Fast Fourier Transform to identify the k dominant spectral components that contribute the most to the original data. This approach enables the identification of the primary periodic features within the data.

$$\mathbf{A} = \frac{1}{N} \sum_{i=1}^N \left| \text{FFT} \left(\mathbf{X}^{(l-1)} \right)_i \right|, \quad (9)$$

$$\{f_i\}_k = \arg \text{TopK}(\mathbf{A}), \quad p_i = \frac{T}{f_i}. \quad (10)$$

Here, $\mathbf{A} \in \mathbb{R}^T$ indicates the amplitude values, averaged along the d_{model} dimension. Based on the derived period lengths p , the inputs are reshaped into 3D tensors using the following equations, yielding multiple representations corresponding to different time scales.

$$\mathbf{X}_{\text{patch}}^i = \text{Reshape}_{p_i}(\text{Padding}(\mathbf{X}^{(l-1)})) \quad (11)$$

where $\text{Padding}()$ applies to zero-padding along the temporal dimension T , ensuring alignment with the requirements of $\text{Reshape}()$. $\mathbf{X}_{\text{patch}}^i$ denotes the reshaped time series corresponding to the i -th scale.

At each scale, multi-head attention is employed to capture local relationships within the series. Specifically, for each scale $\mathbf{X}_{\text{patch}}^i$, multi-head attention is applied along the period dimension using the following equation:

$$\mathbf{H}^i = \text{Multi-Head Attention}(\mathbf{X}_{\text{patch}}^i) \quad (12)$$

Finally, to transition to the next layer, the feature tensors across different scales are combined through a weighted summation, with the weights determined by the amplitude associated with each scale. This process is formally expressed by the following formula:

$$\alpha_1, \dots, \alpha_k = \text{SoftMax}(\mathbf{A}_{f_1}, \dots, \mathbf{A}_{f_k}), \quad (13)$$

$$\mathbf{H}_{\text{out}} = \sum_{i=1}^k \alpha_i \mathbf{H}^i. \quad (14)$$

Here, $\mathbf{A}_{f_1}, \dots, \mathbf{A}_{f_k}$ represent the amplitudes of each scale, and $\mathbf{H}_{\text{out}}$ represents the output from the current multi-scale attention layer. This aggregation strategy enables the model to weight information from each scale according to the amplitudes, effectively fusing features across multiple scales.

In brief, the module proceeds through four stages. First, the input series is mapped into a latent feature space. Second, the module utilizes FFT to capture the dominant periodic structures in the data. Third, the series is reshaped according to these dominant periods, and multi-head attention is applied to capture local dependencies specific to each temporal scale. Finally, the features extracted from different scales are dynamically aggregated using weights derived from their respective frequency amplitudes. The underlying intuition behind the workflow is to allow the model to adaptively capture and integrate local patterns across varying temporal granularities, ensuring that critical features at different scales are effectively fused.

C. Frequency Branch

A frequency-domain branch equipped with complex-domain attention is designed to capture global patterns arising from spectral combinations. In practice, abnormal behaviors occurring in the temporal domain are often accompanied by shifts in spectral energy allocation. While amplitude represents the strength of periodic patterns, phase encodes the critical temporal positioning (i.e., when a pattern occurs). Contextual anomalies, such as phase shifts or rhythm mismatches, drastically alter this complex-valued coupling. Therefore, by directly applying a complex-valued multi-head attention mechanism along the

Algorithm 1: Training Process for TFT-GCN.

Input: Original time series X , values n , layers N
Output: Reconstructed data $\hat{X}$
while $t < \text{MaxEpoch}$ **do**
 initialize $\text{vec}_1, \text{vec}_2$
 $E \leftarrow \text{Compute_adjacency_matrix}(\text{vec}_1, \text{vec}_2)$
 for $i = 1$ **to** D **do**
 $\mathcal{N}_{\text{neg}}(v_i) \leftarrow \text{Bottom-N}_{\text{neg}}(E(v_i))$
 $\mathcal{N}_{\text{pos}}(v_i) \leftarrow \text{Top-N}_{\text{pos}}(E(v_i))$
 $E \leftarrow \text{Re_normalize}(\mathcal{N}_{\text{neg}}, \mathcal{N}_{\text{pos}})$
 $X_g \leftarrow \text{GCN}(E, X)$
 $X^0 \leftarrow X_g$
 $F^0 \leftarrow \text{Fourier_Transform}(X)$
 for $l = 1$ **to** N **do**
 $X^l \leftarrow \text{Multi-scale Attention}(X^{(l-1)})$
 $F^l \leftarrow \text{Complex-values Attention}(F^{(l-1)})$
 $\hat{X} \leftarrow X^N + F^N$
 $\mathcal{L} \leftarrow \lambda_1 \mathcal{L}(X, \hat{X}) + \lambda_2 \mathcal{L}_{\text{fft}}(X, \hat{X})$
 $\text{Optimizer}(\mathcal{L})$
 end while

frequency components, our model learns the normal, stable interaction patterns between different frequencies (encompassing both amplitude and phase). During the inference phase, any anomaly that breaks this inherent periodic rhythm will fail to align with the learned normal spectral distribution, resulting in a significantly magnified reconstruction error.

This branch first applies Fourier transform to the time series, converting the input from the temporal to the spectral domain and producing its frequency representation $\mathbf{F} \in \mathbb{C}^{\hat{L} \times D}$. Notably, a channel-independent transformation strategy is adopted, as the temporal-domain branch already accounts for inter-variable relationships. The frequency-domain branch primarily addresses the challenge of capturing global dependencies within individual series.

Next, we apply the RevIN strategy to normalize the frequency representation, aligning the spectra of different global feature series to a similar distribution. The matrix F is then transposed and embedded as $\hat{\mathbf{F}} \in \mathbb{C}^{D \times \text{dim}}$ using a linear projection. Given the weak temporal correlations in the frequency domain spectra, positional embedding is deemed unnecessary. Subsequently, a complex-valued multi-head attention mechanism is applied along the dim -dimension. For the h -th attention head, the query, key, and value representations are defined as $Q_h = \hat{\mathbf{F}} W_h^Q, K_h = \hat{\mathbf{F}} W_h^K, V_h = \hat{\mathbf{F}} W_h^V$, where $W_h^Q, W_h^K, W_h^V \in \mathbb{C}^{\text{dim} \times d}$.

$$\text{Att}_h(Q_h, K_h, V_h) = \text{Softmax}(|Q_h K_h^T|) V_h \quad (15)$$

The multi-head attention mechanism generates its final output by combining the results from all heads through concatenation:

$$\text{MHA}(\hat{\mathbf{F}}) = \text{Concat}(\text{Att}_1, \dots, \text{Att}_H)^T W_O \quad (16)$$

where $W_O \in \mathbb{C}^{Hd \times \text{dim}}$ is a learnable matrix.

The frequency-domain branch incorporates the Feed Forward layer and residual connections from the Transformer architecture. After multiple layers of processing, the final output $\hat{\mathbf{F}}_O$ is mapped back to $\mathbf{F}_O \in \mathbb{C}^{\hat{L} \times D}$. The final step involves an inverse Fourier operation, restoring the representation to the time domain and yielding the frequency branch output $\mathbf{X}_{\text{freq}} \in \mathbb{R}^{L \times D}$.

D. Output and Optimization

The final output of TFT-GCN is obtained by summing the outputs from the temporal and frequency branches. Although this fusion strategy is structurally simple, it is designed based on three considerations. Firstly, the temporal branch mainly captures local fluctuations and cross-variable dependencies, while the frequency branch focuses on global periodic patterns. Late fusion helps preserve the complementary characteristics of the two domains and avoids premature feature interference. Secondly, compared with complex cross-domain interaction modules (e.g., cross-attention), simple summation introduces little additional computational overhead and maintains the efficiency of the model. Thirdly, although the two branches are decoupled in structure, they are jointly optimized during end-to-end training, which allows the temporal and frequency representations to be implicitly coordinated through shared optimization objectives.

The training objective aims to preserve the intrinsic characteristics of the observed sequence during reconstruction. Additionally, Fons et al. [28] demonstrated that incorporating a Fourier-based loss term can enhance reconstruction fidelity for time series data. Consequently, the training loss is defined as:

$$\mathcal{L} = \lambda_1 \|x - \hat{x}\|^2 + \lambda_2 \|\text{FFT}(x) - \text{FFT}(\hat{x})\|^2 \quad (17)$$

where $\hat{x}$ denotes the reconstructed version of x . During inference, anomaly scores are computed according to the reconstruction discrepancy measured in the MSE sense.

IV. EXPERIMENTS AND ANALYSIS

This section assesses the proposed method's effectiveness through detection performance, ablation studies, computational complexity analysis, hyperparameter impact analysis, and a discussion of limitations.

A. Benchmark Datasets

Experiments were conducted on seven widely used benchmark datasets to evaluate the proposed framework. The following is a brief description of each dataset.

- **MSL [4] & SMAP [4]:** The MSL and SMAP datasets, publicly available and expert-labeled by NASA, include multivariate monitoring signals collected during rover and satellite operations. Specifically, MSL involves 27 monitored objects with 55 measurement channels, whereas SMAP contains 55 monitored objects and 25 channels. Notably, the SMAP dataset contains a higher proportion of point anomalies compared to MSL.

TABLE I

OVERALL RESULTS ON SEVEN BENCHMARK DATASETS. ANOMALY* INDICATES ANOMALY TRANSFORMER. THE OPTIMAL RESULTS ARE MARKED IN BOLD, AND THE SECOND-BEST RESULTS ARE INDICATED IN UNDERLINED.

Method	SMAP			SMD			SWaT			MSL			PSM			HAI			SKAB			Average F1 %
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1	
LSTM-VAE	0.8645	0.5673	0.6851	0.7576	0.9008	0.823	0.76	0.895	0.822	0.8549	0.7994	0.8262	0.7362	0.8992	0.8096	0.786	0.851	0.8173	0.953	0.931	0.9424	81.79
DLinear	0.8987	0.5383	0.6733	0.7589	0.8397	0.7973	0.9226	0.9306	0.9266	0.8968	0.753	0.8187	0.9865	0.9468	0.9662	0.794	0.8494	0.8208	0.9684	0.927	0.9483	85.02
Autoformer	0.9081	0.625	0.7404	0.7845	0.6511	0.7116	0.9996	0.6556	0.7918	0.9054	0.7472	0.8187	0.9999	0.7895	0.8823	0.7936	0.7103	0.7496	0.9663	0.8855	0.9241	80.26
Informer	0.8967	0.5607	0.6899	0.7833	0.6521	0.7117	0.9981	0.6733	0.8042	0.9012	0.7369	0.8108	0.992	0.831	0.9044	0.788	0.7126	0.7484	0.9865	0.9149	0.9493	80.27
FEDformer	0.9017	0.5548	0.6869	0.728	0.8157	0.7693	0.9995	0.6556	0.7918	0.9065	0.7524	0.8223	0.9999	0.8189	0.9004	0.7915	0.7108	0.7490	0.9654	0.891	0.9267	80.66
FuSAGNet	0.9071	0.5762	0.6993	0.7868	0.8237	0.8046	0.9878	0.726	0.8369	0.8955	0.7529	0.818	0.9852	0.9183	0.9489	0.8679	0.7479	0.8034	0.965	0.931	0.9474	83.69
Anomaly*	0.9185	0.5811	<u>0.7118</u>	0.8891	0.8223	<u>0.8549</u>	0.7251	0.9732	0.831	0.7708	0.8941	<u>0.8278</u>	0.6835	0.9472	0.794	0.7934	0.7144	0.7519	0.9805	0.9149	0.9466	81.69
iTransformer	0.9067	0.53	0.669	0.7629	0.8529	0.8054	0.9221	0.9306	0.9263	0.8616	0.6261	0.7252	0.9792	0.918	0.9476	0.7617	0.8395	0.7987	0.9765	0.9273	<u>0.9513</u>	83.19
PatchTST	0.8662	0.5673	0.6856	0.7846	0.8809	0.8299	0.9096	0.7973	0.8489	0.8881	0.7285	0.8004	0.9881	0.9265	0.9563	0.8312	0.827	0.8286	0.9778	0.9234	0.9498	84.28
TimesNet	0.9002	0.5647	0.694	0.8789	0.8154	0.8459	0.9214	0.931	0.9262	0.8955	0.7529	0.818	0.9852	0.9623	0.9736	0.8089	0.8663	0.8368	0.9767	0.9262	0.9507	<u>86.36</u>
GPT4TS	0.9008	0.5573	0.6886	0.8729	0.8082	0.8393	0.9212	0.9308	0.9259	0.8182	0.813	0.8156	0.9835	0.9582	<u>0.9707</u>	0.857	0.831	<u>0.8433</u>	0.962	0.911	0.9348	85.97
ModernTCN	0.9083	0.5593	0.6923	0.8743	0.8164	0.8444	0.9577	0.9029	<u>0.9295</u>	0.8958	0.7491	0.8159	0.9866	0.9446	0.9652	0.9281	0.7128	0.812	0.958	0.937	0.9468	85.8
TFT-GCN	0.9039	0.5596	0.6913	0.8869	0.8452	0.8655	0.9333	0.9535	0.9433	0.9025	0.7791	0.8363	0.9756	0.962	0.9687	0.7948	0.9061	0.8468	0.9812	0.9303	0.9551	87.24

- *PSM* [29]: The PSM dataset, publicly available from eBay, includes 25 features collected from multiple internal application server nodes.
- *SMD* [16]: The SMD, collected over five weeks from an Internet company's computing cluster, records statistics from 28 machines involving 33 variables.
- *SWaT* [30]: The SWaT dataset is generated from a compact water treatment testbed operated by Singapore's Public Utility Board, spanning 11 days of operation with both normal and attack scenarios.
- *HAI* [31]: The HAI dataset originates from an industrial control system (ICS) testbed integrated with a Hardware-In-the-Loop (HIL) simulator for steam-turbine and pumped-storage hydropower generation.
- *SKAB* [32]: The Skoltech Anomaly Benchmark is a public multivariate time series benchmark for anomaly detection in industrial control systems (ICS).

B. Baselines

We compare twelve methods for anomaly detection, including transformer-based approaches such as Autoformer [7], FEDformer [8], Informer [33], iTransformer [23], Anomaly Transformer [12], and PatchTST [9]. The comparison further involves LSTM-VAE [34], DLinear [35], and the graph-based method FuSAGNet [13]. For a more comprehensive evaluation, we also include a large model-based method GPT4TS [36], along with the convolution-based models TimesNet [10] and ModernTCN [27].

C. Experimental Setups

All experiments were implemented with PyTorch and executed on an NVIDIA RTX 4080 platform. For every dataset, the validation subset was constructed from the trailing 20% segment of the training data in accordance with commonly adopted experimental settings. Specifically, the training and testing splits are as follows: MSL with 53,817 (training) and 73,729 (testing) samples of dimension 55, SMAP with 135,183 and 427,617 samples of dimension 25, SMD with 708,405 and 708,420 samples of dimension 38, SWaT with 495,000 and 449,919 samples of dimension 51, PSM with 132,481 and 87,841 samples of dimension 25, HAI with 309600 and 291600 samples of

dimension 59, and SKAB with 1147 and 1145 samples of dimension 8. The model includes two encoder layers, with 8 attention heads (H) for both temporal and frequency attention. The parameters d_{model} and $fnet_{d_{model}}$ are tuned individually for each dataset. The optimization process was performed using Adam with an initial step size of 1×10^{-3} . The training configuration used 5 epochs, a sequence window of length 100, and mini-batches containing 32 samples. Baseline methods adhered to their respective hyperparameter configurations.

D. Evaluation Metrics

The model's performance is evaluated using F1-score, precision (P) and recall (R):

$$F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad P = \frac{TP}{TP + FP} \quad R = \frac{TP}{TP + FN} \quad (18)$$

Consistent with prior anomaly detection studies on time series data, the point-adjustment (PA) strategy is adopted for evaluation. Under this criterion, an anomalous interval is considered successfully identified once any timestamp within that interval is detected, while points outside anomaly segments are evaluated normally.

E. Main Comparative Results

To assess the performance and generalization of TFT-GCN, we compared it against several leading anomaly detection methods on seven widely-used datasets. Table I reports the comparative results on seven benchmark datasets, where TFT-GCN attains the best F1 performance on most datasets. Specifically, TFT-GCN shows improvements of 1.23% ($0.8549 \rightarrow 0.8655$), 1.48% ($0.9295 \rightarrow 0.9433$), and 1.03% ($0.8278 \rightarrow 0.8363$) on the SMD, SWaT, and MSL datasets, respectively. Moreover, on the HAI and SKAB datasets, TFT-GCN also achieves consistent improvements over the baseline methods, although the gains are less than 1%. This indicates that our model maintains stable and robust performance across diverse datasets with different characteristics. Additionally, we assessed generalization by averaging F1 scores across all datasets, with TFT-GCN consistently outperforming other models in terms of its average ranking. However, TFT-GCN's performance on the SMAP and

PSM datasets is suboptimal, which can be attributed to the inherent limitations of its components when handling specific data characteristics. For SMAP, Autoformer [7] exhibits superior performance. As SMAP contains a high proportion of point anomalies (i.e., sharp, isolated spikes), these anomalies manifest as high-frequency transient signals. Our frequency-domain branch, designed to emphasize global periodic trends, naturally attenuates high-frequency variations. Consequently, it may inadvertently over-smooth these critical spikes, making it less sensitive than Autoformer's time-domain mechanism. Regarding PSM, TimesNet [10] gains a slight advantage. The PSM dataset comprises IT server metrics, which are loosely coupled and highly dynamic. Unlike industrial control systems with strict physical constraints, PSM lacks stable, long-term inter-variable dependencies. Because our cross-variable GCN is optimized to learn invariant structural relationships, it struggles to construct a robust graph in such environments, whereas TimesNet effectively models 2D temporal variations without relying on fixed variable topologies.

TFT-GCN significantly outperforms other models on the highly non-stationary SMD and SWaT datasets, demonstrating its ability to overcome the limitations of previous methods on such data. The strength of TFT-GCN lies in its dual analysis of time series, utilizing the temporal and spectral perspectives. Within the temporal framework, the cross-variable GCN captures complex interactions between variables, while the multi-scale attention module extracts short-range patterns under different temporal granularities. In the spectral branch, the model learns spectral features and global patterns by analyzing frequency components. The complementary nature of the temporal and spectral analyses allows TFT-GCN to better integrate local and global information, improving its performance on complex datasets.

F. Evaluation Under Strict Point-Wise Protocol

As point-adjustment (PA) is widely adopted in previous studies, the main results are conducted under PA for fair comparison with existing methods. However, recent works point out that PA may overestimate performance by treating an entire anomalous segment as correctly detected once any point is identified. To further validate the genuine detection capability of TFT-GCN, we additionally conduct strict point-wise evaluation without any point-adjustment and compare against the strongest baselines, ModernTCN and TimesNet. Quantitative comparisons are summarized in Table II.

Although all models exhibit noticeable performance degradation under the stricter metric, TFT-GCN remains the top-performing approach on four of five datasets (SMD, SWaT, MSL, and PSM), demonstrating that its superiority does not rely on PA inflation. These results indicate that the proposed temporal multi-scale attention and frequency-domain modeling improve the model's ability to accurately identify anomalous points under strict evaluation settings.

G. Model Analysis

1) *Ablation Studies*: To assess the impact of individual model components, we systematically removed individual

TABLE II
PERFORMANCE COMPARISON UNDER STRICTLY UNADJUSTED (RAW)
POINT-WISE EVALUATION WITHOUT POINT-ADJUSTMENT

Method	Metric	SMAP	SMD	SWaT	MSL	PSM
TFT-GCN	P	0.2055	0.3639	0.2728	0.2308	0.5787
	R	0.0159	0.0692	0.0318	0.0406	0.0308
	F1	0.0296	0.1163	0.0572	0.0690	0.0585
ModernTCN	P	0.2965	0.3394	0.3741	0.2506	0.6082
	R	0.0239	0.0623	0.0249	0.0291	0.0227
	F1	0.0442	0.1053	0.0467	0.0522	0.0438
TimesNet	P	0.2679	0.3589	0.2474	0.2412	0.6532
	R	0.0229	0.0627	0.0263	0.0299	0.0280
	F1	<u>0.0422</u>	<u>0.1067</u>	<u>0.0475</u>	<u>0.0531</u>	<u>0.0537</u>

components and examined the resulting changes in model performance. Specifically, we assessed the roles of spectral attention (frequency branch), the time branch, multi-scale attention, and cross-variable GCN. The variants of TFT-GCN are described below:

- i) *w/o-FB*: We removed the frequency branch.
- ii) *w/o-TB*: We removed the temporal branch.
- iii) *w/o-TA*: We removed multi-scale attention of the temporal branch.
- iv) *w/o-GCN*: We removed cross-variable GCN in the temporal branch and eliminated the inter-series correlation learning.
- v) *w/o-Floss*: We removed the Fourier loss from the loss function.

The ablation results are summarized in Table III. Our analysis of these results is summarized as follows:

- i) *Improvement of frequency branch*: The model's overall effectiveness significantly decreases after removing the frequency-domain branch, indicating that learning spectral features and capturing global patterns are crucial for detecting anomalies. To validate the conclusion, we integrate the frequency-domain attention mechanism into the TimesNet [10] model and compare the test performance of the original TimesNet and modified TimesNet model across different datasets, as shown in Fig. 3. The modified model, incorporating frequency-domain attention, outperforms the original TimesNet by 1.09% (0.8459 → 0.8551) on the SMD dataset, 0.93% (0.9262 → 0.9348) on the SWaT dataset, and 1.16% (0.818 → 0.8275) on the MSL dataset. These results further demonstrate that frequency-domain feature analysis significantly enhances time series anomaly detection.
- ii) *Improvement of time branch*: The w/o-TB experiments reveal that analyzing only the frequency components while neglecting temporal variations still significantly impacts anomaly detection.
- iii) *Improvement of multi-scale attention*: The results from w/o-TA clearly show that multi-scale attention contributes to the improvement of model performance, suggesting the existence of local features at different scales.

TABLE III
ABLATION ANALYSIS ON FIVE BENCHMARK DATASETS: PRECISION, RECALL,
AND F1 SCORES

Method	Metric	SMAP	SMD	SWaT	MSL	PSM
TFT-GCN	P	0.9039	0.8869	0.9333	0.9025	0.9756
	R	0.5596	0.8452	0.9535	0.7791	0.962
	F1	0.6913	0.8655	0.9433	0.8363	0.9687
w/o-FB	P	0.8987	0.7838	0.9107	0.8834	0.9613
	R	0.5578	0.8794	0.9562	0.7358	0.9706
	F1	0.6883	0.8289	0.9329	0.8029	0.9659
w/o-TB	P	0.8589	0.8753	0.9302	0.8941	0.9728
	R	0.5678	0.8236	0.9296	0.7708	0.944
	F1	0.6836	0.8487	0.9299	0.8278	0.9582
w/o-TA	P	0.8974	0.8849	0.941	0.9008	0.9802
	R	0.5544	0.8118	0.9291	0.7757	0.9235
	F1	0.6854	0.8468	<u>0.935</u>	<u>0.8336</u>	0.951
w/o-GCN	P	0.9059	0.8878	0.8989	0.8984	0.974
	R	0.5518	0.8334	0.9549	0.7639	0.9485
	F1	0.6859	<u>0.8597</u>	0.926	0.8257	0.9611
w/o-Floss	P	0.899	0.871	0.9248	0.8991	0.967
	R	0.5541	0.839	0.9367	0.77	0.9485
	F1	0.6857	0.854	0.9307	0.8296	0.955

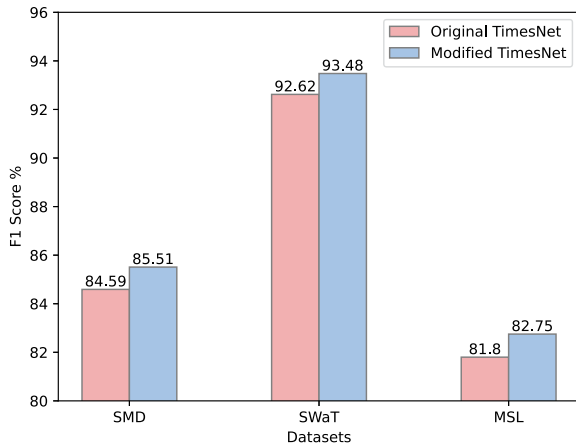


Fig. 3. Performance comparison between original TimesNet and modified TimesNet on different datasets.

- iv) *Improvement of cross-variable GCN*: The w/o-GCN results demonstrate that the cross-variable GCN structure improves model performance, as w/o-GCN slightly underperforms compared to TFT-GCN. This indicates that the dependency relationships between series contribute to improved outcomes. To better highlight the advantages of the cross-variable GCN in capturing complex inter-series relationships, we replace it with a standard GCN and compare performance before and after the modification in Fig. 4. The modified model performs worse across four

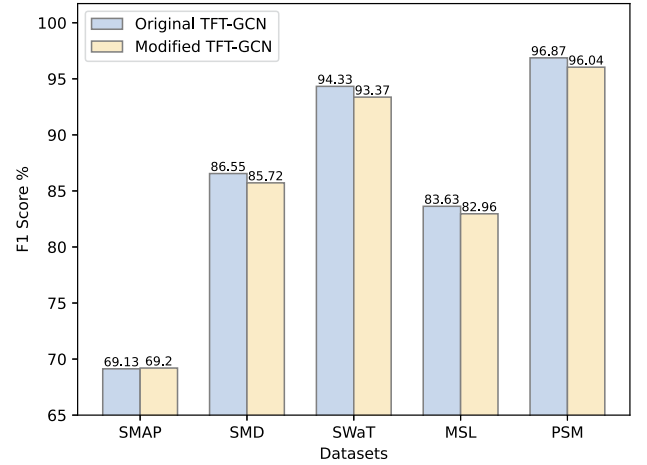


Fig. 4. Performance comparison of TFT-GCN with homo-heterogeneous GCN and standard GCN.

datasets, except for the SMAP dataset, where performance remains unchanged. This further confirms that the cross-variable GCN effectively learns invariant associations, incorporating both homogeneous and heterogeneous dependencies, which boosts the model's anomaly detection capability.

- v) *Improvement of Fourier loss*: The results from w/o-Floss demonstrate the significance of Fourier loss in the model training process.

2) *Visual Analysis*: We demonstrate the effectiveness of TFT-GCN by visualizing anomalies in Fig. 5. To highlight its advantages, we compare TFT-GCN with TimesNet [10], a classical model in time series analysis. Specifically, we visualize the anomalies and anomaly scores for both models. Fig. 5(a) and (b) respectively illustrate the detection performance of TimesNet and TFT-GCN for regional anomalies and point anomalies. Evidently, TFT-GCN demonstrates superior capability in detecting outliers by establishing more appropriate anomaly scores to distinguish them from normal data points.

3) *Computational Complexity*: To evaluate the practical efficiency and the accuracy-efficiency tradeoff of the proposed model, we report multiple empirical metrics, including training time, floating-point operations (FLOPs), and inference time across different benchmark datasets. As summarized in Table IV, we compare TFT-GCN with two representative baselines: TimesNet (representing high accuracy) and ModernTCN (representing high efficiency). The results reveal that TimesNet incurs massive computational overhead—reaching Giga-level FLOPs on datasets like SMAP and SMD—due to its complex 2D-variation transformations. In contrast, TFT-GCN drastically reduces both FLOPs and time consumption to the Mega-level, while concurrently achieving superior detection accuracy (as demonstrated in Table I). Although ModernTCN requires the least computational resources owing to its lightweight pure-convolutional structure, its anomaly detection capability falls short of our method in complex scenarios. Overall, TFT-GCN strikes an excellent balance, delivering robust anomaly detection

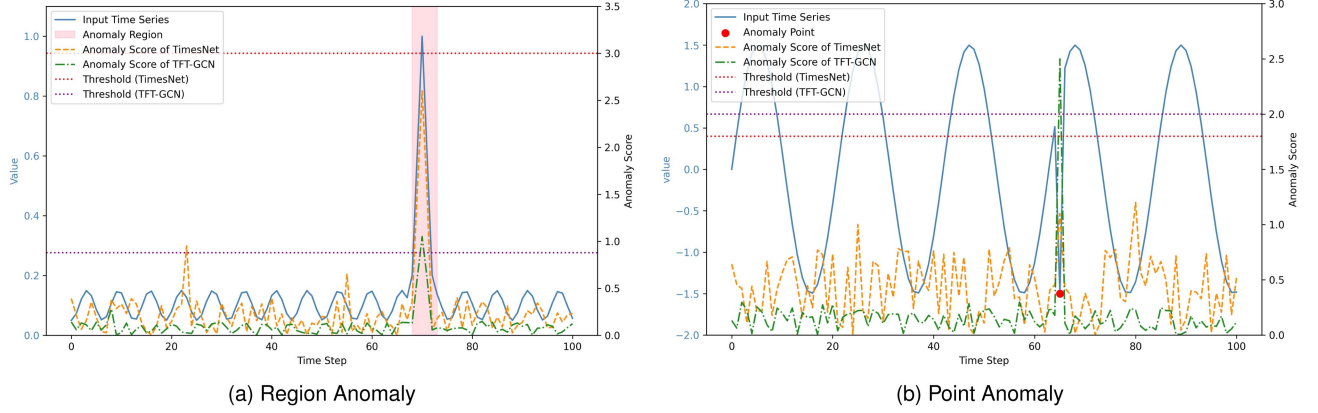


Fig. 5. Comparisons of ground-truth anomaly and anomaly score between TFT-GCN and TimesNet.

TABLE IV
THE TRAINING TIME(TT), FLOPS AND INFERENCE TIME(IT) ON FIVE DATASETS

Method	Metric	SMAP	SMD	SWaT	MSL	PSM
TFT-GCN	TT	22.41m	1.01m	84.18m	12.38m	22m
	FLOPS	0.68G	0.37G	0.28G	0.15G	0.37G
	IT	3.98m	4.61s	5.5m	1.33m	46.61s
ModernTCN	TT	13.78m	10.72s	54.79m	1.05m	14.5m
	FLOPS	0.10G	7.42M	0.21G	2.22M	0.21G
	IT	1.21m	0.15s	1.30m	1.49s	22.44s
TimesNet	TT	200m	59.89s	132.26m	2.04m	7.3m
	FLOPS	9.34G	2.34G	33.19M	22.29M	1.56G
	IT	40m	2.4s	1.16m	4.16s	15.17

with highly acceptable computational demands for real-world applications.

Regarding the computational cost, the multi-scale attention module in the temporal branch contributes most of the overall complexity. Since multi-head attention is performed across k reshaped temporal scales, its worst-case complexity is $\mathcal{O}(k \cdot T^2 \cdot d_{\text{model}})$. In contrast, the cross-variable GCN introduces relatively low overhead, with complexity bounded by $\mathcal{O}(T \cdot D^2)$. As the number of variables D is usually much smaller than the sequence length T , the graph operation remains efficient. In the frequency branch, FFT requires only $\mathcal{O}(T \log T)$ operations, and the subsequent complex-valued attention is applied on the embedded feature dimension rather than directly along the temporal dimension, avoiding the quadratic temporal bottleneck.

Despite integrating multiple modules, TFT-GCN maintains stable training behavior. First, Reversible Instance Normalization (RevIN) alleviates distribution shifts in non-stationary time series, providing stable inputs for subsequent layers. Second, residual connections and Layer Normalization are adopted in both branches to improve optimization stability. Finally, the dual-domain reconstruction objective jointly constrains the temporal and frequency representations, helping reduce overfitting to noise in a single domain and promoting stable convergence.

4) *Parameter Sensitivity*: We further investigated the effect of parameter variations on TFT-GCN. As depicted in Fig. 6(a), the performance variation of TFT-GCN is systematically evaluated under different configurations of encoder layers, while maintaining a consistent architecture in the time-frequency branch. The number of network layers is a critical hyperparameter in deep neural networks; too many layers may cause training challenges, overfitting, and degradation, while too few layers may result in insufficient model expressiveness, preventing the model from learning complex patterns effectively. As seen in Fig. 6(a), model performance varies across different datasets as encoder layers increase. For several datasets, the performance improves with increasing layer depth, whereas for others, performance stabilizes or slightly declines when exceeding a critical layer threshold. Overall, the TFT-GCN architecture maintains robust performance across varying encoder depths, achieving optimal performance metrics at a configuration of 2 encoder layers. Therefore, in the subsequent evaluations, we set the encoder layers to a fixed number of 2. As shown in Fig. 6(b), the sensitivity of the TFT-GCN to different attention head configurations is analyzed. Experiments reveal consistent performance enhancement with the incremental increase in attention heads, peaking at 8 heads across most datasets. This suggests that an increased number of attention heads improves the model's capacity to effectively extract a wide range of feature variations. Fig. 6(c) and (d) illustrate the effect of frequency domain attention dimension $fnet_{d_{\text{model}}}$ and the value of n in cross-variable GCN on the model performance, respectively. It can be observed that the impact of attention dimension and n -value of GCN on model performance varies across datasets, which may be attributed to the differences in the feature dimensions and the relationships across the variables. Fig. 6(e) shows the effect of the K value on model performance in the Fourier transform of the time-domain branch. It can be seen that the impact of the K value is consistent across all three datasets.

5) *Discussion of Limitations*: While TFT-GCN demonstrates strong overall performance, several limitations remain. First, under conditions of extreme anomaly sparsity, the model may face difficulties in learning discriminative representations due to the lack of sufficient abnormal signals. Second, when

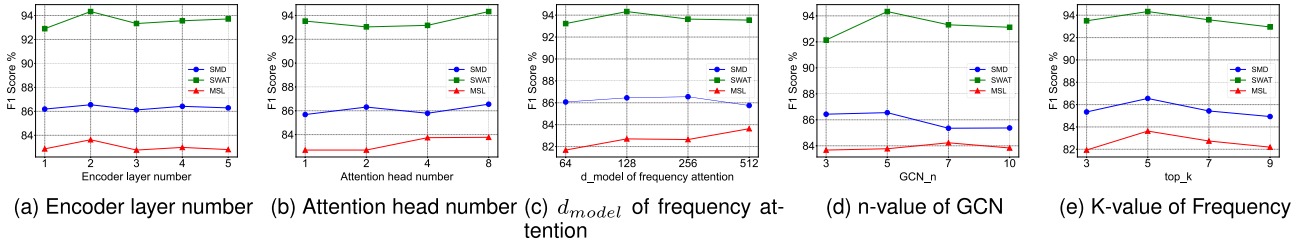


Fig. 6. Sensitivity analysis of key hyperparameters in TFT-GCN.

dealing with irregularly sampled sequences, temporal dependencies and frequency-domain representations may be distorted, which could adversely affect detection accuracy. Such issues are broadly observed in sequential anomaly analysis rather than being specific to TFT-GCN alone. Future research will focus on data augmentation strategies to better capture rare anomalies and design model architectures that can explicitly accommodate irregular sampling patterns.

V. CONCLUSION

This paper proposes the TFT-GCN model, which combines temporal and spectral information to improve the accuracy of detecting anomalies. The approach successfully captures global trends and local features within the data. A distinctive characteristic of TFT-GCN is the incorporation of an attention mechanism in the frequency domain, which models the interrelationships between the frequency components, thereby enhancing the recognition of global patterns. Additionally, TFT-GCN employs a cross-variable graph convolutional network to efficiently address both homogeneous and heterogeneous relationships among time series variables. Extensive evaluations conducted on diverse datasets reveal that TFT-GCN achieves superior overall performance compared to several advanced methods. Our findings underscore the importance of analyzing time series from multiple perspectives. Future work will focus on addressing challenges such as extreme anomaly sparsity and irregular sampling, by integrating data augmentation strategies and developing architectures capable of handling unevenly distributed temporal information.

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